

Digital Forensics Case Report

Windows Host-Based Investigation

This case involves the forensic analysis of a compromised Windows system after the execution of a malicious attack script. The goal of this investigation is to perform host-based forensic triage, analyze key Windows Registry artifacts, identify user actions that led to the attack, reconstruct the execution chain, and determine persistence mechanisms on the system.

The investigation was conducted as a series of structured laboratory tasks, each focusing on specific artifacts and forensic techniques commonly used in incident response.

Scope of Investigation

The investigation focused on:

- Rapid triage collection using KAPE
- Registry analysis of SYSTEM, SOFTWARE, SAM, and NTUSER.DAT hives
- Identification of executed scripts and binaries
- Correlation of user activity leading to attack execution
- Detection of persistence mechanisms
- Documentation of findings and remediation actions

Tools Used

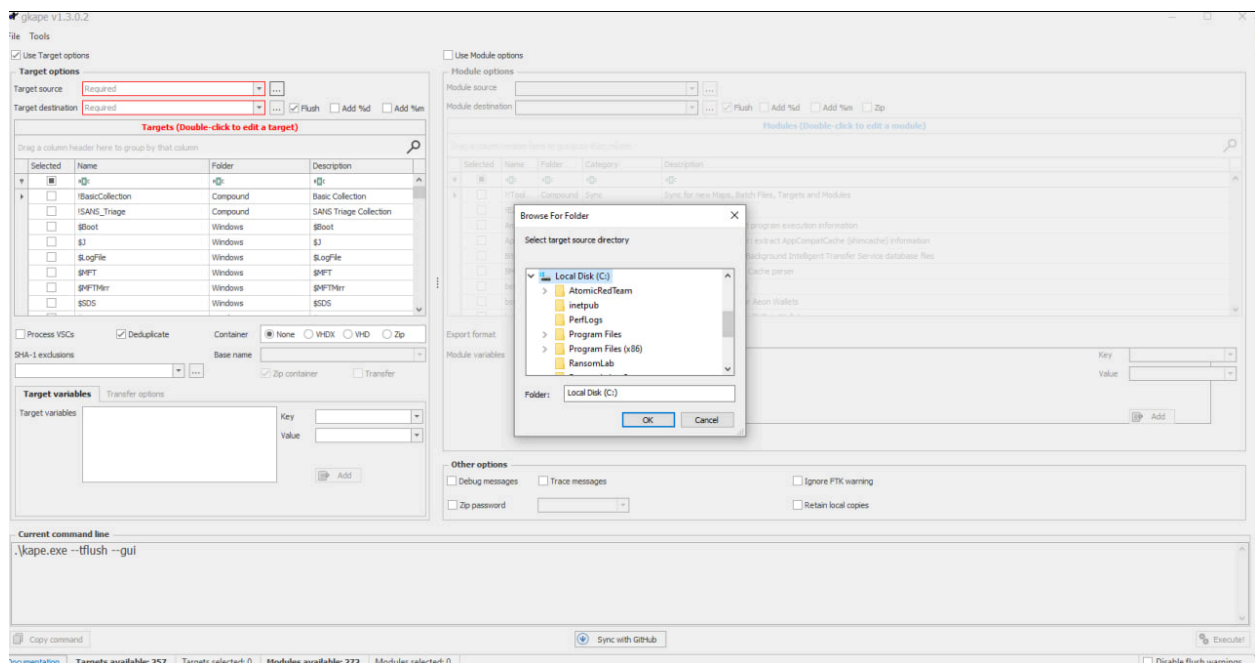
The following tools were used throughout the investigation:

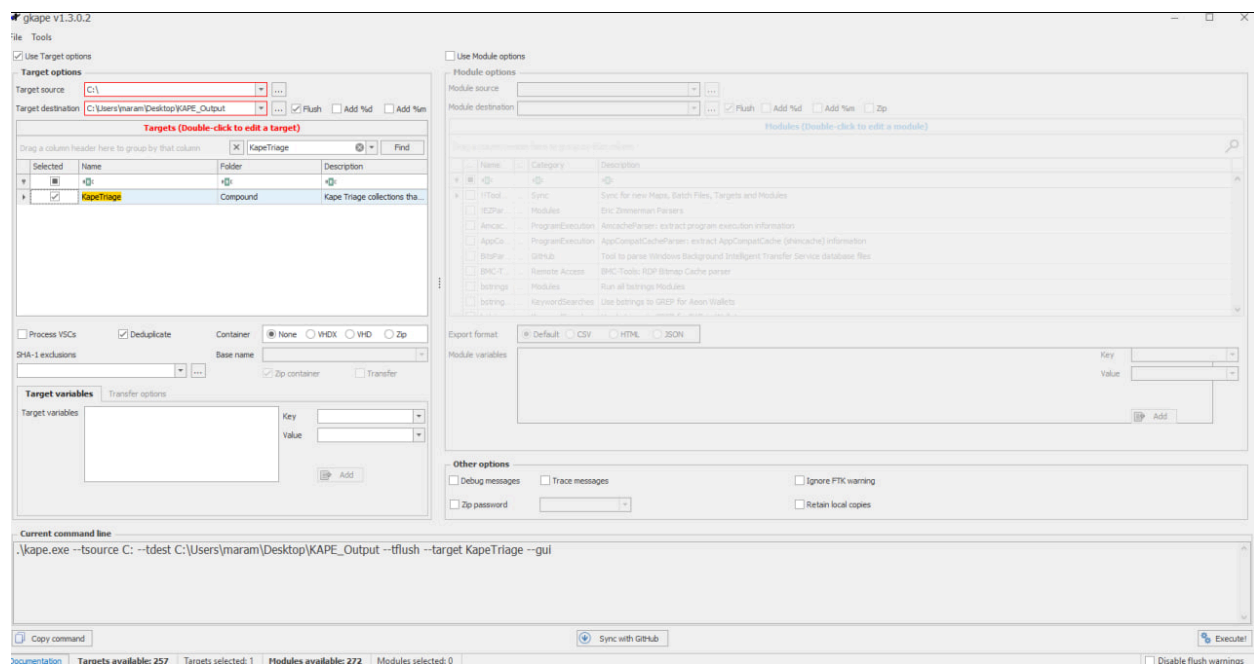
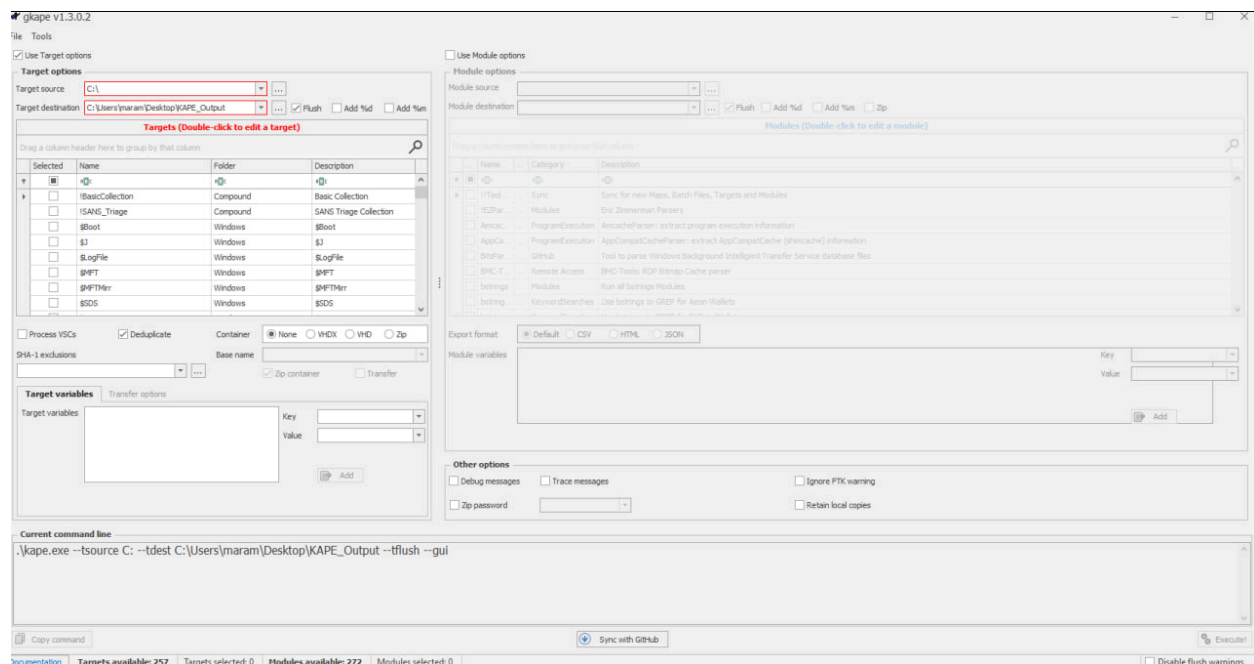
- **KAPE (Kroll Artifact Parser and Extractor)** – for triage collection
- **Registry Explorer** – for offline registry analysis
- **PECmd** – for Prefetch artifact analysis
- **Event Viewer / Windows PowerShell.evtx** – for PowerShell and execution activity

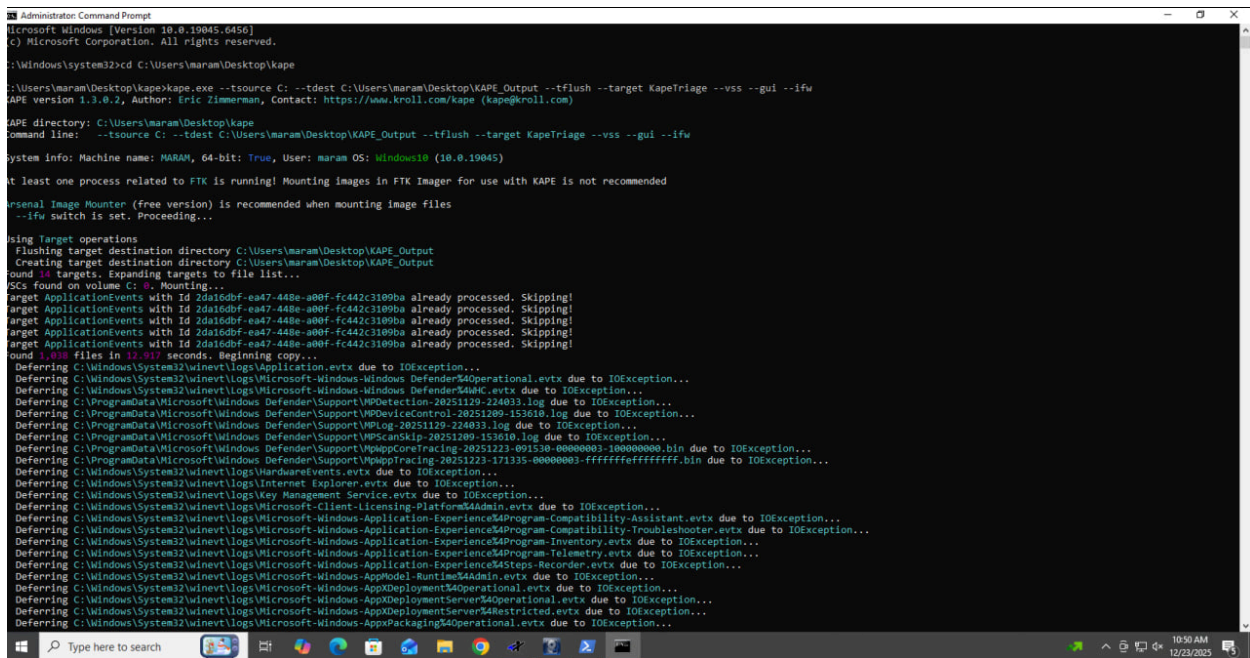
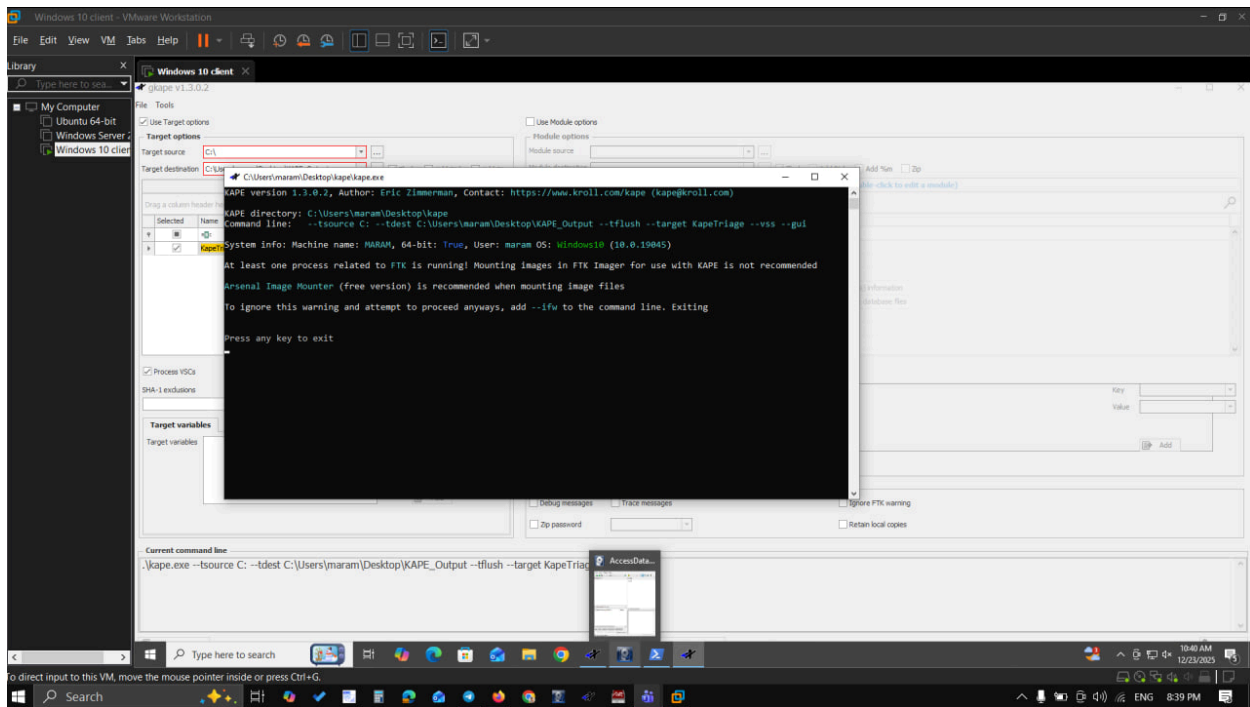
- **Manual File System Analysis** – for user profiles and persistence validation

1) TRIAGE PACKAGE CREATION

- KAPE configuration
- Source and destination paths
- Selected targets/modules
- Successful triage execution







2) MACHINE INFORMATION (SYSTEM & SOFTWARE HIVES)

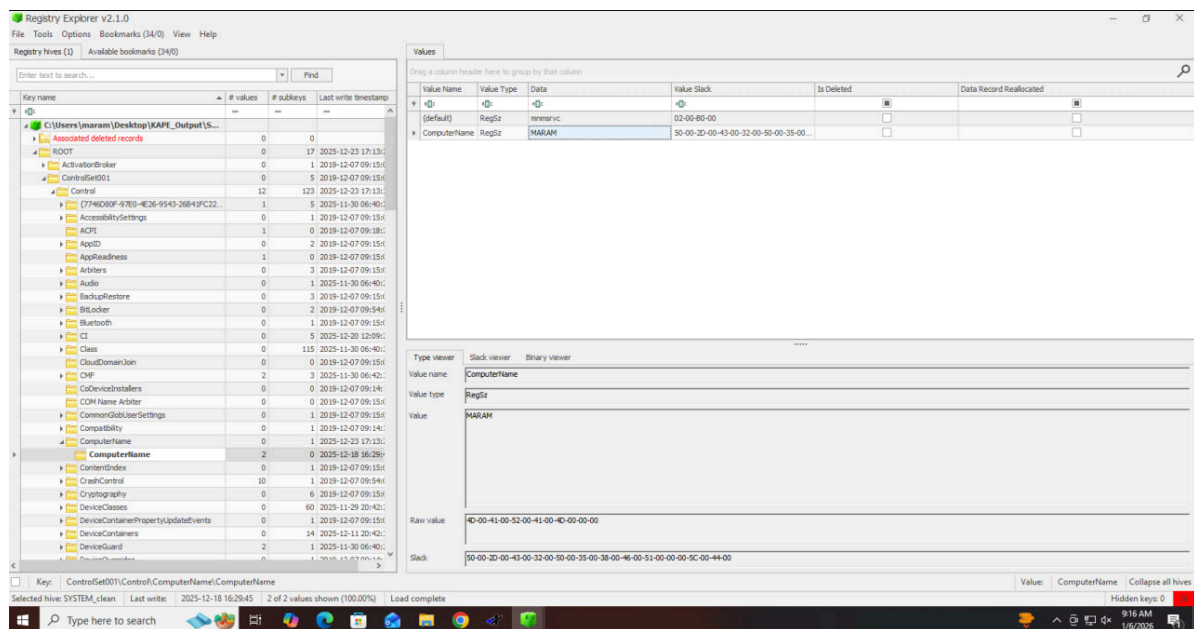
Findings

1. Machine Name

The computer name was identified from the **SYSTEM** registry hive.

Registry Path: `HKLM\SYSTEM\CONTROLSET001\COMPUTERNAME\COMPUTERNAME`

- **Value Data:** `MARAM`

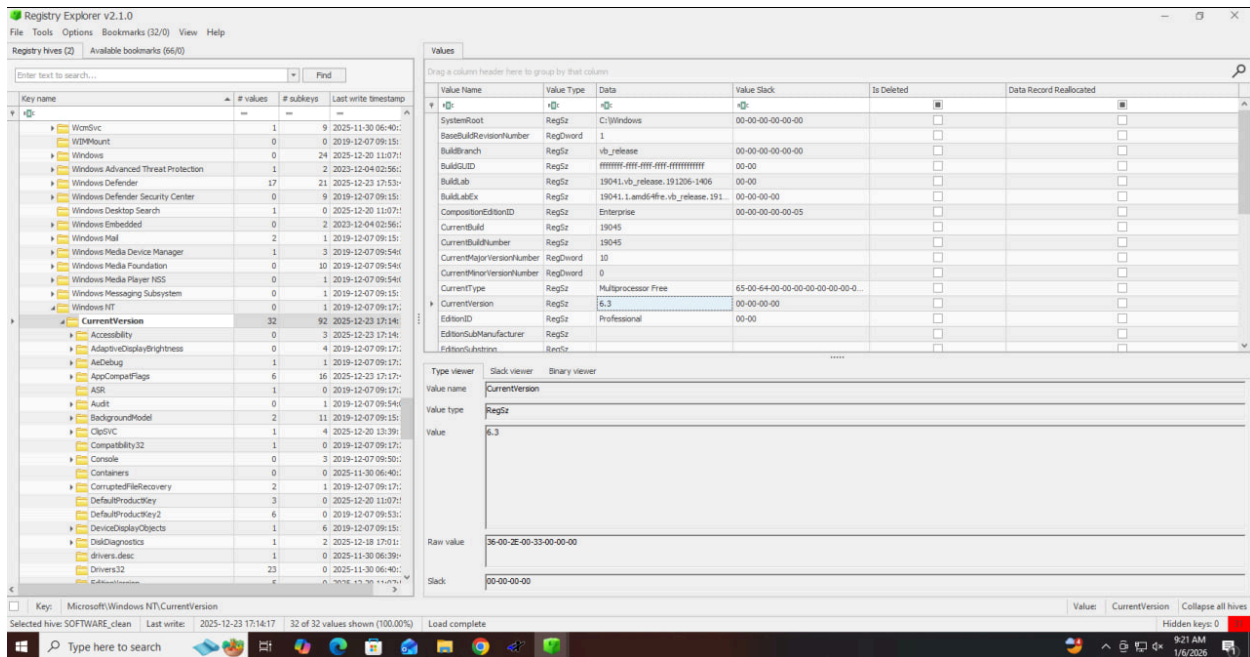


2. Windows Version

Windows version information was extracted from the SOFTWARE registry hive.

Registry Path: `HKLM\SOFTWARE\Microsoft\WindowsNT\CURRENT VERSION`

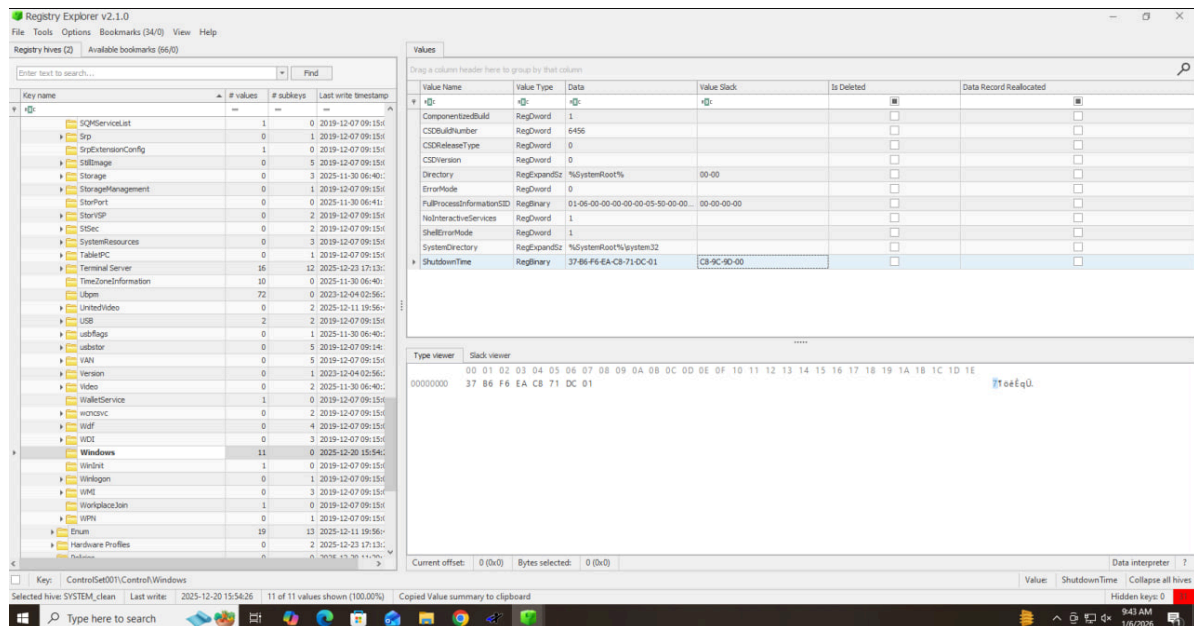
- **Value Data:** `6.3`



3. Last Shutdown Time

The last system shutdown information was identified from the SYSTEM registry hive.

- **Registry Path:**
- **HKLM\SYSTEM\ControlSet001\Control\Windows**



3) SAM HIVE ANALYSIS

Analyze the SAM hive to identify local accounts and detect newly created or suspicious accounts.

Artifacts Examined

- SAM hive
- Users directory on disk

The SAM hive was examined to enumerate local users. Account creation times, group memberships, and profile existence were validated to identify potentially malicious accounts.

Extracted Account Details

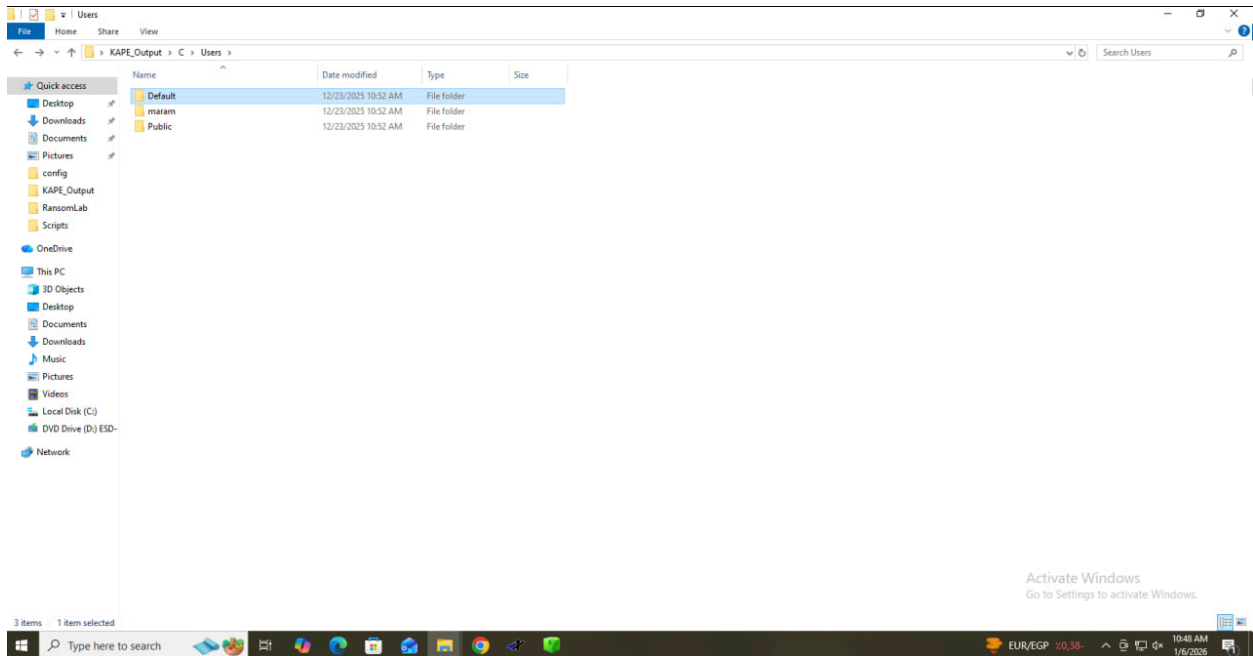
- Account Name
- Account Creation Time
- Group Membership

Registry Path: **SAM\SYSTEM\Domains\Accounts\Users**

The screenshot displays the Registry Explorer v2.1.0 interface. The left pane shows the tree structure of the registry, with the path **C:\Users\maram\Desktop\KAPE_Output\Users** selected. The right pane shows a table of user accounts extracted from the SAM hive. The table has columns for Valid, User, Inval, Total, Crea, Last, Last, Expr, Us, Full, Pass, Groups, Com, User, Hom, Inter, Rese, Acco, Hom, Pass, Tem, Norm, Mne, Inter, Work, Serv, Pass, and Auto. The table contains 6 rows of data, including accounts like Administrator, System, Guest, and maram. Below the table, the 'Type viewer' section shows the value name (default), value type (RegDwordBigEndian), and value (0). The status bar at the bottom indicates the selected hive is SAM, last written on 2025-12-23 17:47:21, and 1 of 1 values shown (100.00%).

Valid	User	Inval	Total	Crea	Last	Last	Expr	Us	Full	Pass	Groups	Com	User	Hom	Inter	Rese	Acco	Hom	Pass	Tem	Norm	Mne	Inter	Work	Serv	Pass	Auto
	Administrator		1002	0	2025-12-23 17:47:21			Administrators			Administrators, Users																
	System		503	0	2025-12-23 17:47:21			Default (Account)			System, Administrators, Users																
	Guest		501	0	2025-12-23 17:47:21			Guest			Guests																
	maram		1001	0	2025-12-23 17:47:21			maram			Administrators																

- **User Profile Presence (C:\Users)**



The account had been created but was not used in the attack , so it does not have profile because it was not logged in with.

4) NTUSER.DAT ANALYSIS

Identify user actions that led to execution of the attack script via NTUSER.DAT hive analysis.

Artifacts Examined

- NTUSER.DAT registry hive

The NTUSER.DAT hive was analyzed to reconstruct user actions preceding attack execution. Registry artifacts related to script execution and user interactions were correlated to establish the execution chain.

Identified Evidence

- **Registry path associated with executed script**

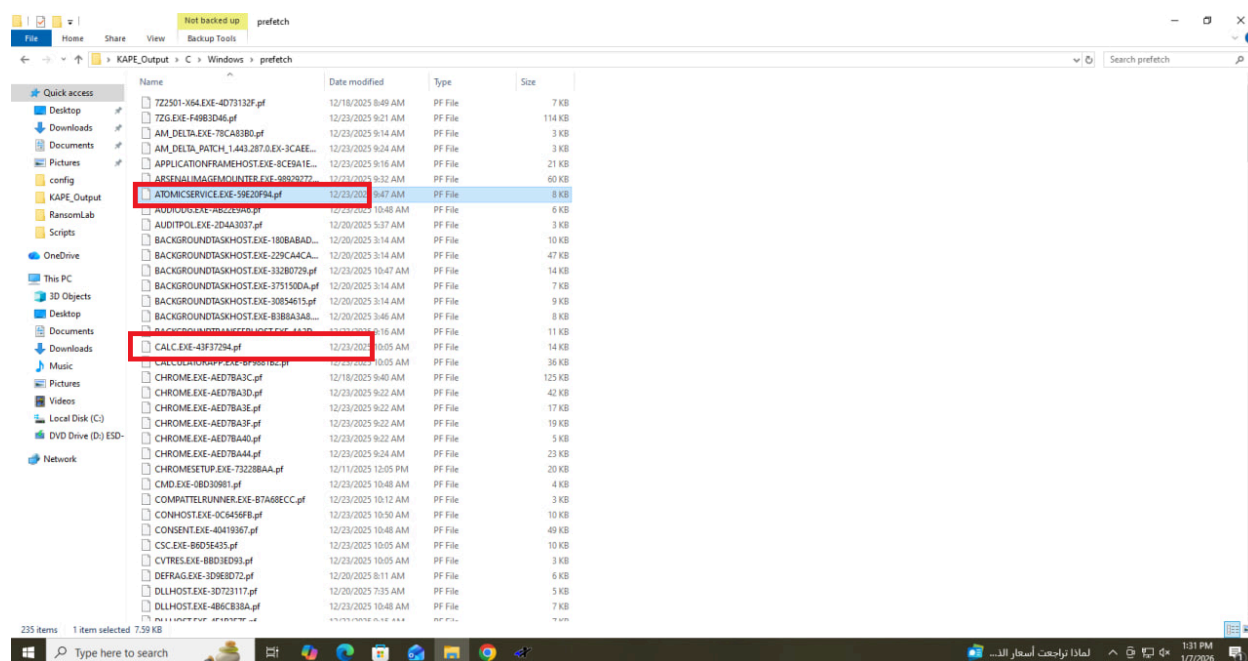
Registry Path: `SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs\ps1`

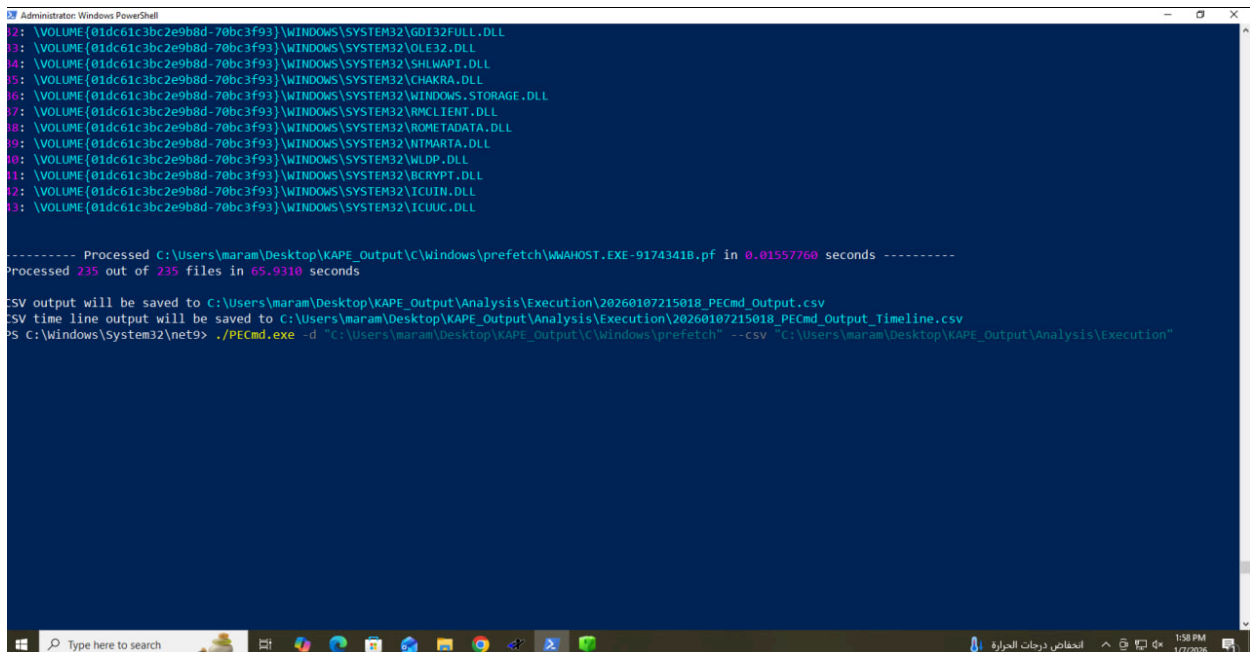
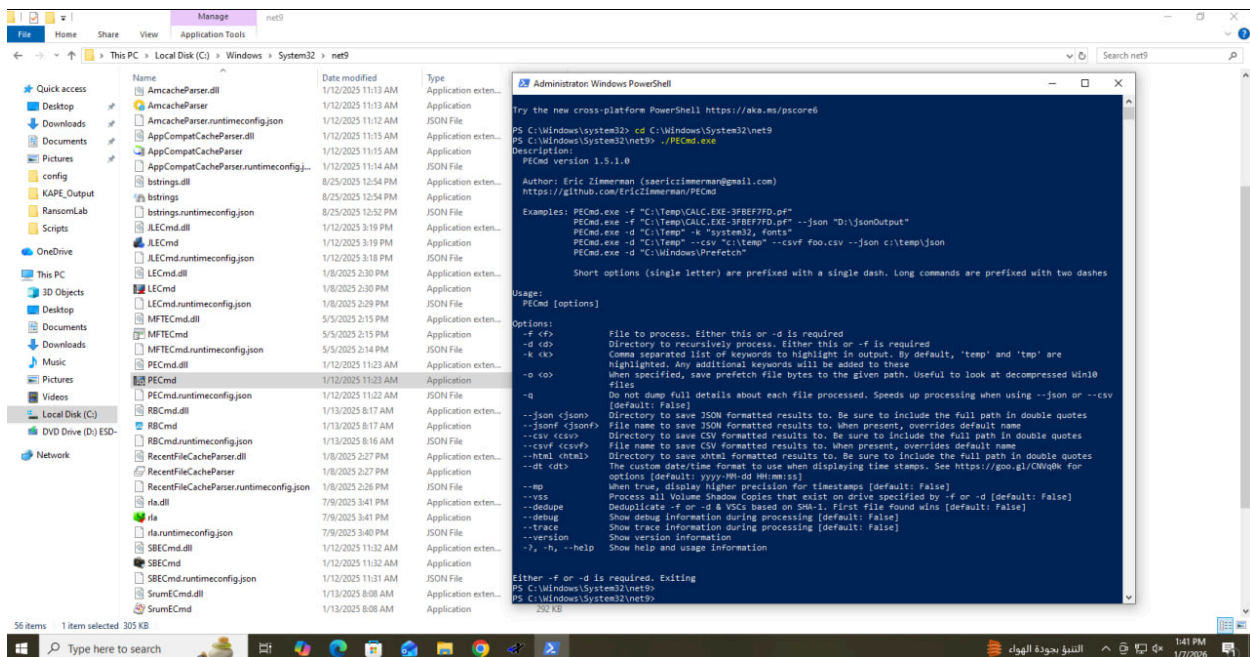
Identify execution behavior observed during the attack, including evidence of PowerShell usage, calculator execution, registry interactions, and service activity.

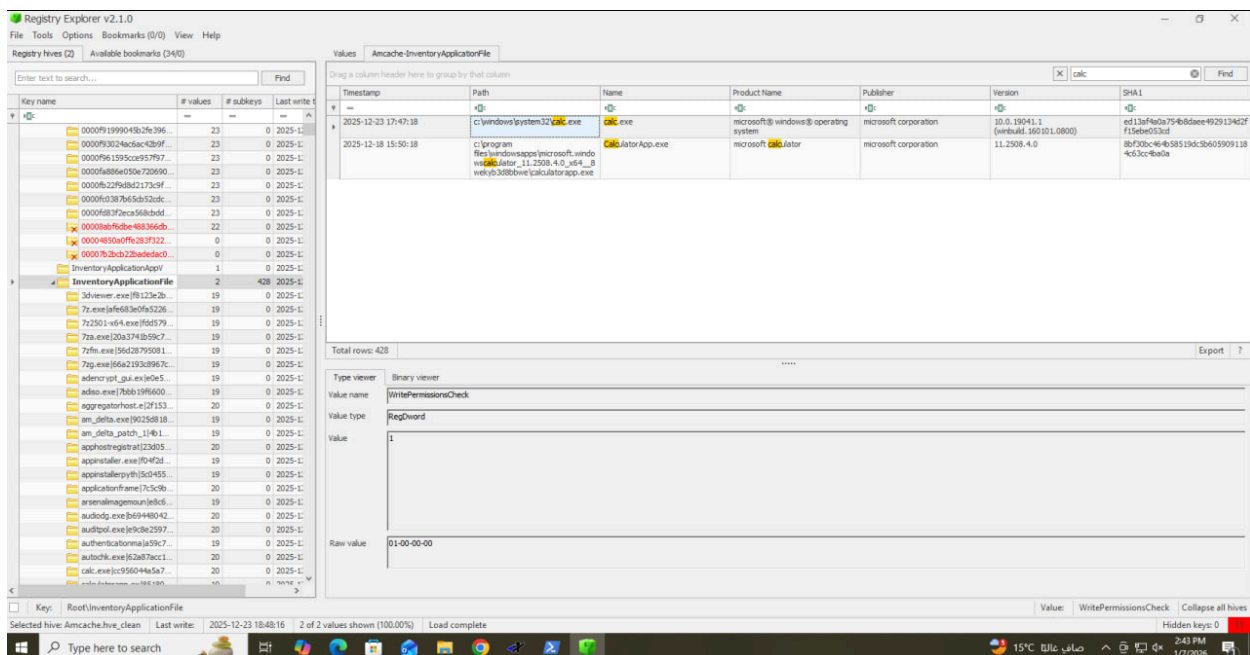
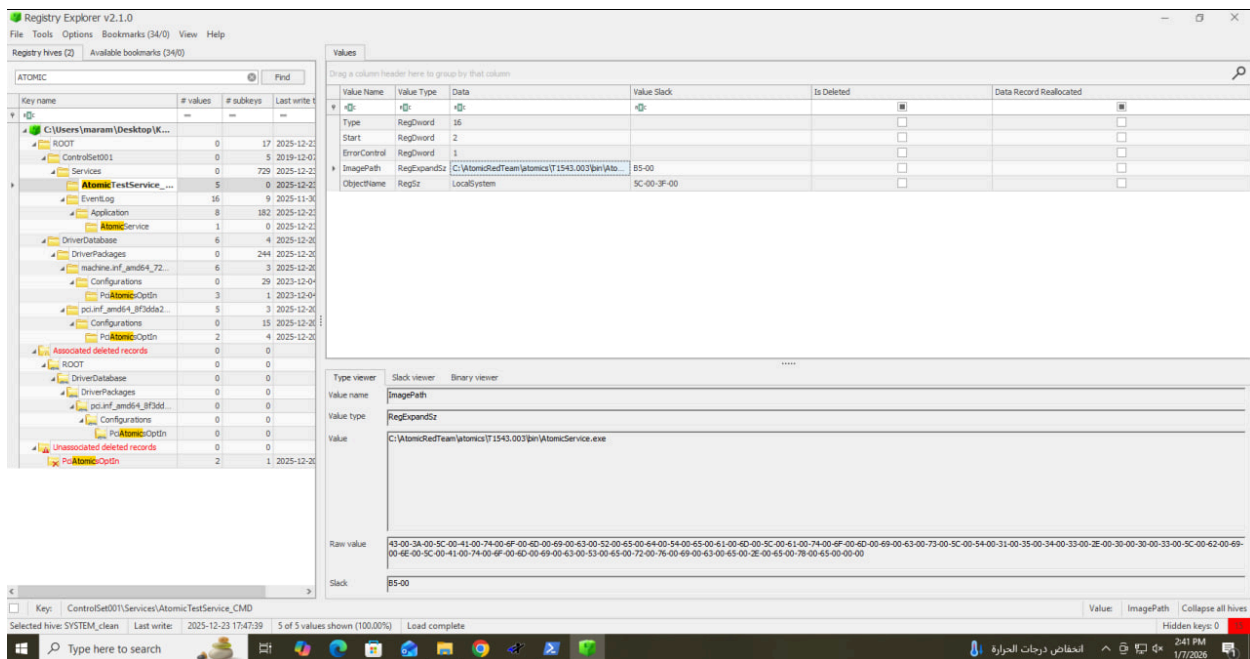
Artifacts Examined

- Prefetch files (PECmd)
- Registry-related process artifacts
- Service execution artifacts

Execution artifacts were analyzed to confirm behaviors during the attack. PowerShell activity was correlated with registry modifications and secondary process execution, including Calculator. Service-related artifacts were analyzed for links to Atomic Red Team activity







Based on the analyzed artifacts, the execution chain can be summarized as follows:

1. PowerShell was executed on the system.
2. Atomic Red Team PowerShell scripts were launched.
3. A child process (Calculator) was spawned during script execution.

This confirms that PowerShell served as the primary execution vector, leading to successful process execution on the system.

6) Persistence Mechanism

Persistence Mechanism #1 – User-Level Registry Run Key

A persistence mechanism was identified within the current user's registry hive (**NTUSER.DAT**). The attacker created a registry value under the *Run* key, causing a malicious executable to be launched automatically each time the user logs in.

Artifact Details

- **Registry Hive:** NTUSER.DAT
- **Registry Path:**

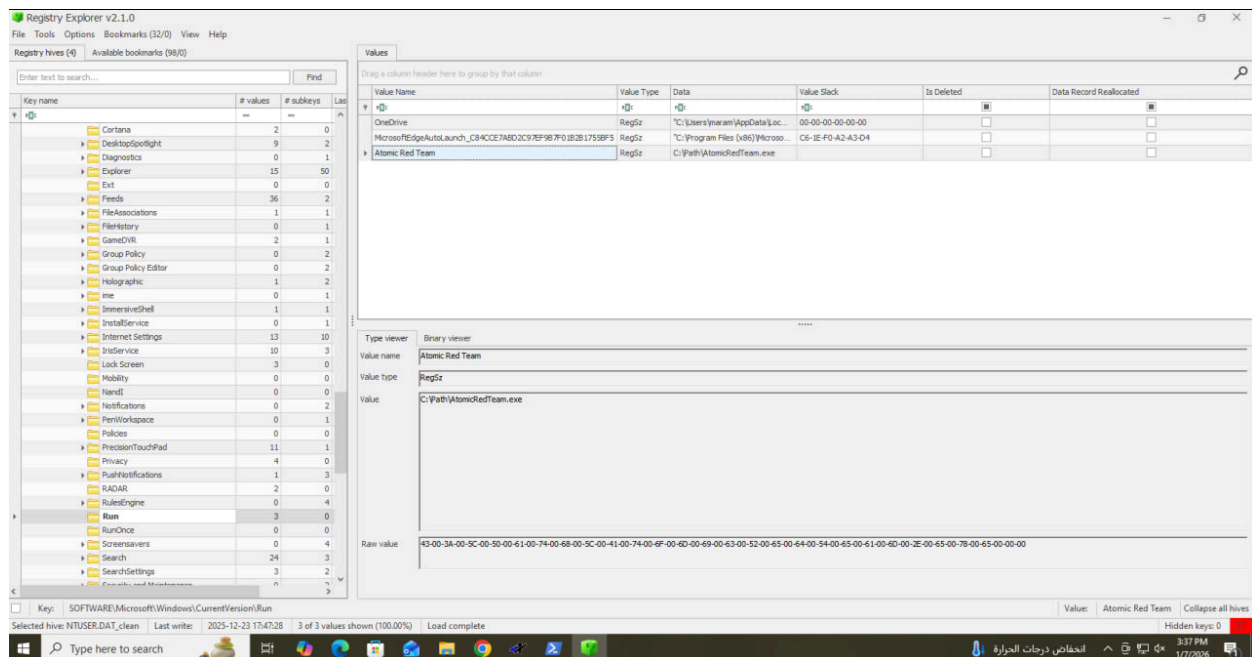
SOFTWARE\Microsoft\Windows\CurrentVersion\Run

- **Value Name:** Atomic Red Team
- **Value Type:** REG_SZ
- **Data:**

C:\Path\AtomicRedTeam.exe

- **Last Write Time:** 2025-12-23 17:47:28

This confirms the presence of **user-level persistence**, ensuring execution of the malicious payload upon user logon.



Persistence Mechanism #2 – Malicious Windows Service

A second persistence mechanism was identified in the **SYSTEM** registry hive, where a malicious Windows service was created. Services configured with an automatic start mode execute during system startup, allowing persistence even before any user logs in.

Artifact Details

- **Registry Hive:** SYSTEM
- **Registry Path:**

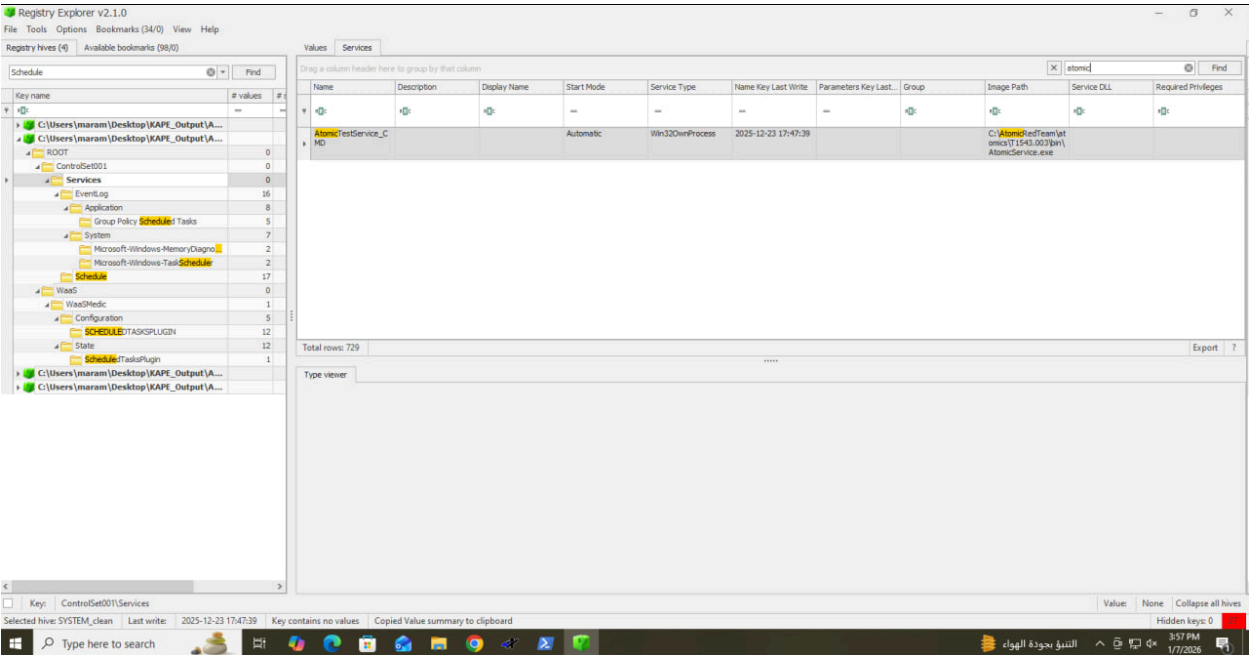
ControlSet001\Services\AtomicTestService_C

- **Service Name:** AtomicTestService_C
- **Start Mode:** Automatic
- **Service Type:** Win32OwnProcess
- **Image Path:**

C:\AtomicRedTeam\atomics\T1543.003\bin\AtomicService.exe

- **Last Write Time:** 2025-12-23 17:47:39
- This confirms **system-level persistence** via a malicious service, ensuring execution on every system boot.

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Remediation Actions

- Removal of malicious registry entries
- Deletion of associated executables or scripts
- Validation of system startup and scheduled tasks
- Verification that no unauthorized services remain active

MITRE ATT&CK Technique Execution Results

T1078.003 – Valid Accounts: Local Account Creation

T1059.001 – Command and Scripting Interpreter: PowerShell (Fileless Execution)

T1547.001 – Boot or Logon Autostart Execution: Registry Run Key

T1053.005 – Scheduled Task / Job: Scheduled Task on Logon

T1543.003 – Create or Modify System Process: Windows Service

Conclusion

The investigation successfully reconstructed the attack timeline, identified the execution chain, and uncovered multiple persistence mechanisms. Using systematic registry and artifact analysis, the compromised system was remediated and restored to a secure state.